■■ FakeShield AI — Complete Project Workflow & System Architecture

AI-Powered Unified Scam, Phishing & Digital Threat Detection Platform

1. Executive Platform Architecture

FakeShield AI is an enterprise-level cybersecurity threat detection platform designed to protect users against multi-vector digital threats including phishing URLs, fraudulent emails, scam SMS/WhatsApp messages, dangerous QR code links, image payment screenshots (via OCR), and malicious PDF/TXT documents.

Step	Phase / Module	Workflow Mechanism & Processing Details
1	**Input Ingestion**	User submits URL, Email text, SMS/WhatsApp message, QR Code image, Screenshot image, or PDF document.
2	**Scanner Pipeline**	Dedicated scanner module validates input, extracts features, performs OCR, or decodes QR matrix using OpenCV QRCodeDetector.
3	**ML & NLP Engine**	Runs 14-Feature Random Forest URL Classifier or TF-IDF Logistic Regression text scam classifier.
4	**Threat Intel**	Queries VirusTotal v3 API & performs WHOIS/SSL/Subdomain structural analysis.
5	**AI Risk Engine**	Calculates unified 0–100 Risk Score (■ SAFE 0-25, ■ LOW 26-50, ■ MEDIUM 51-75, ■ HIGH 76-100).
6	**XAI & Multilingual**	Generates Explainable AI (XAI) reasonings and preventive action steps in English, Marathi (■■■■■), and Hindi (■■■■■).
7	**Output & Reports**	Renders interactive gauge result, updates dashboard charts, saves to history, and exports PDF/Excel reports.

2. Deep Vector Scanner Breakdown

■ **URL Scanner:** Extracts 14 features (URL length, dots, hyphens, special chars, digits, HTTPS, IP host, suspicious keywords, @ symbol, shortening service, subdomains, domain age, SSL, redirects) and passes vector to Random Forest ML model.

■ **Email &** ■ **Message Scanner:** Uses TF-IDF vectorization + Logistic Regression + 8 Scam Rules (Banking, OTP, Job, Lottery, Investment, KYC, Delivery, UPI Scam).

■ **QR Code Scanner (Fixed & Upgraded):** Uses OpenCV *cv2.QRCodeDetector* + PIL image decoders to read QR code image pixels directly and decode payload URLs.

■■ **Image OCR &** ■ **Document Scanner:** Extracts embedded text from screenshots or parses PDF/DOCX content to detect fraudulent financial requests.

3. Explainable AI (XAI) & Multilingual Action Plan

FakeShield AI does not simply output a binary prediction. It explains **WHY** content is dangerous through XAI bullet reasons, and provides actionable safety recommendations in three languages:

English: Do not click unverified links or share passwords/OTPs with anyone.

[illegible] (Marathi [illegible]): [illegible] Link [illegible] [illegible] [illegible] [illegible] [illegible] OTP / Password [illegible] [illegible] [illegible] [illegible].

[illegible] (Hindi [illegible]): [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]

4. Resolution of QR Code Scanning Issue

Problem Resolved: The QR Code scanner has been upgraded to use OpenCV's native cv2.QRCodeDetector() engine. This ensures that uploaded PNG, JPG, JPEG, and BMP QR images are accurately read directly from image pixel arrays without requiring external zbar C-libraries, eliminating decoding failures.